

cc4714/1027341
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Supervised setting

$$y \in \{0, 1\}$$

$$x \in \mathcal{R}^n$$

$$h_0 \in \mathcal{H} \rightarrow \text{finite}$$

All possible instances



$$\{x^{(1)}, y^{(1)}, \dots, x^{(m)}, y^{(m)}\}$$

$$x^{(i)}, y^{(i)} \sim \text{Dist.}$$

$$(x, y) \sim \text{Dist}$$

$$\hat{\epsilon}_0(h_0) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{h_0(x^{(i)}) \neq y^{(i)}\}$$

Average training error

$$\arg \min_{h_0} \hat{\epsilon}_0(h_0)$$

$$\frac{1}{m} \sum_{i=1}^m \mathbb{1}\{h_0(x^{(i)}) \neq y^{(i)}\}$$

$$\epsilon(h_0) = E_{(x,y) \sim \text{Dist}} [\mathbb{1}\{h_0(x) \neq y\}]$$

Generalization error

$$\Pr(|\hat{\epsilon}_0(h_0) - \epsilon(h_0)| > \gamma) \leq \delta \quad (1)$$

PAC: - Probably Approximately Correct $\downarrow$ const

$$\gamma, \delta, m, |\mathcal{H}| = K$$

$\downarrow$ # of examples

$$\rightarrow \Pr(\exists h_0 \in \mathcal{H}: |\hat{\epsilon}_0(h_0) - \epsilon(h_0)| > \gamma) \leq \delta'$$

Coin Toss Experiment

$$\{x^{(i)}\}_{i=1}^m$$

$$x \in \{0, 1\}$$

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$$\hat{\phi} = \frac{1}{m} \sum_{i=1}^m x^{(i)}$$

$$x^{(1)} \dots x^{(m)}$$

ϕ : - Probability

$$x=1$$

$$x^{(i)} \sim \text{Bernoulli}(\phi)$$

$$\Pr(x=1)$$

$$E[\hat{\phi}] = \frac{1}{m} \sum_{i=1}^m E[x^{(i)}]$$

Trial:

$$\begin{matrix} \#H & \#T \\ 47 & 53 \\ 52 & 48 \end{matrix}$$

$$x^{(i)} \sim \text{Bernoulli}(\phi) \Rightarrow \frac{1}{m} \sum_{i=1}^m x^{(i)} = \phi$$

Unbiased estimator

$$\text{Var}[\hat{\phi}] = \frac{1}{m^2} \sum_{i=1}^m \text{Var}[x^{(i)}]$$

$$= \frac{1}{m} \phi(1-\phi) = \frac{1}{m} \phi(1-\phi)$$

Hoeffding Inequality

Bound -

$$\Pr(|\hat{\phi} - \phi| > \gamma) \leq 2e^{-2\gamma^2 m}$$

$$\text{Our problem: } \{x^{(i)}, y^{(i)}\}_{i=1}^m$$

$$z^{(i)} = \{ \frac{1}{2} \mathbb{1}_{\{y^{(i)} \neq y^{(i)}\}} \}$$

$$\hat{z}^{(i)} = \frac{1}{m} \sum_{i=1}^m z^{(i)}$$

$$P \left(\left| \hat{z}^{(i)} - z^{(i)} \right| > r \right) \leq 2e^{-2r^2 m}$$

$$z^{(i)} = E_{(x,y) \sim \text{Dist}} [\mathbb{1}_{\{x \neq y\}}]$$

$$z^{(i)} \sim \text{Bernoulli}(p^{(i)})$$

$$z = \{ \mathbb{1}_{\{x \neq y\}} \}$$

$\hookrightarrow$ Random Variable
(x,y) ~ Dist